

Mathias Jackermeier

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EDUCATION

PhD in Machine Learning – University of Oxford Topic: Deep Reinforcement Learning AIMS Centre for Doctoral Training EPSRC funding	Oct '22–Feb '27
MSc in Computer Science – University of Oxford Focus: Machine Learning Passed with distinction Full scholarship from DAAD	Oct '20–Sep '22
Study abroad – University of Illinois at Urbana-Champaign GPA: 4.0/4.0 Exchange programme with the Department of Computer Science	Aug '18–Dec '18
BSc in Informatics – Technical University of Munich (TUM) Passed with distinction (grade 1.1) Top 2% of cohort	Oct '16–May '20

EXPERIENCE

Student Researcher – Google DeepMind, Frontier AI Unit - Research on efficient, model-driven context management for LLM agents - Developed inference-time and training approaches for tool-based context management - Our method achieves 50-80% peak context reduction while maintaining or improving performance across agentic information retrieval, math reasoning, and coding tasks - Optimised KV cache implementation in Google's Tunix LLM post-training library, reducing memory usage for Gemma4 models by 78% and improving inference speed by 41%	Feb '26–Jul '26
Deep Learning Research Intern – QuantCo Ltd. - Research on LLM agents for completing complex, long-horizon web tasks - Investigated in-context learning from ground-truth demonstrations and SOPs, achieving significant performance improvements across tasks - Extended web benchmarks for long-horizon tasks and developed scalable evaluation infrastructure	Jun '25–Sep '25
PhD Research – Department of Computer Science, University of Oxford - Instruction following and generalisation in multi-task deep RL - Focus on zero-shot generalisation of formally specified tasks (ICLR'25 oral), representation learning (ICML'25 PRAL, IJCAI '26), and multi-task safety guarantees (ICML '26)	Oct '22–present
Stipendiary Lecturer & Teaching Assistant – University of Oxford - Led small-group undergraduate tutorials (St Hugh's College) on Algorithms and Computer Networks - Taught a PhD course on Gaussian processes to first-year students	Oct '24–Mar '25
Software Engineering Intern – CQSE GmbH - Full-stack development of the TeamScale software intelligence platform - Redesigned the frontend and improved performance of rule browser via dynamic loading - Added new language constructs to the TypeScript lexer and parser	Jun '20–Aug '20
Student Research Assistant – Chair for Theoretical Computer Science, TUM - Lead developer of dtControl, a tool for decision tree learning for controller representation - Developed novel algorithms resulting in a 96% size reduction for learned controllers	Jun '19–Jan '20

SKILLS

Languages: Python, Java, C++, JavaScript / TS, SQL, Haskell
ML Frameworks: NumPy, PyTorch, JAX, Tunix

SELECTED PUBLICATIONS

Jackermeier, M., Abate, A. (2025). DeepLTL: Learning to Efficiently Satisfy Complex LTL Specifications for Multi-Task RL. In *ICLR'25 (oral)*, among **top 1.8%** of submissions). <https://deep-ltl.github.io/>.

Schnitzer, Y.*, **Jackermeier, M.***, Abate, A., Parker, D. (2026). Probabilistic Performance Guarantees for Multi-Task Reinforcement Learning. In *ICML'26*. [arXiv:2602.02098](#).

Abate, A., De Giacomo, G., **Jackermeier, M.***, Křetínský, J., Prokop, M.*, Weinhuber, C.* (2026). Semantically Labelled Automata for Multi-Task Reinforcement Learning with LTL Instructions. In *IJCAI'26*. [arXiv:2602.06746](#).

Jackermeier, M., Chen, J., and Horrocks, I. (2024). Dual Box Embeddings for the Description Logic EL++. In *WWW'24 (oral)*. [arXiv:2301.11118](#).